

Joint Online Gaussian Process with Linear Mean Model for Kronecker-Separable Spatio-Temporal HiPPO-SVGP

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1 Joint Observation Model

We consider a spatio-temporal latent response observed at spatial locations $s_i \in \mathcal{S} \subset \mathbb{R}^{d_s}$ and time points $t_n \in \mathbb{R}$. The observation at location (t_n, s_i) is modeled as the sum of a linear mean component, a nonparametric Gaussian process component, and independent Gaussian noise:

$$y(t_n, s_i) = \phi(t_n, s_i)^\top \beta + f(t_n, s_i) + \epsilon_{n,i}, \quad \epsilon_{n,i} \sim \mathcal{N}(0, \sigma^2). \quad (1)$$

Here $\phi(t_n, s_i) \in \mathbb{R}^d$ is a covariate design vector, $\beta \in \mathbb{R}^d$ is the linear mean parameter, and f is a spatio-temporal Gaussian process.

Let the data in the n -th online temporal block be

$$D_n = \{(t, s, y, \phi) : (t, s) \in \mathcal{B}_n\}, \quad (2)$$

and let

$$y_n \in \mathbb{R}^{B_n}, \quad \Phi_n \in \mathbb{R}^{B_n \times d} \quad (3)$$

denote the stacked response vector and design matrix for this block. The block-level observation model is

$$y_n = \Phi_n \beta + f_n + \epsilon_n, \quad \epsilon_n \sim \mathcal{N}(0, \sigma^2 I). \quad (4)$$

The Gaussian process prior is assumed to be separable:

$$f \sim \mathcal{GP}(0, k), \quad k((s, t), (s', t')) = k_s(s, s')k_t(t, t'). \quad (5)$$

If latent values are vectorized over a full time–space grid, then

$$K_{ff} = K_t \otimes K_s. \quad (6)$$

The key modeling decision in this document is to train the linear mean and the GP component jointly. Instead of first estimating β and then fitting a GP to residuals, we directly construct an online posterior over the joint unknowns (β, u) , where u denotes the GP inducing variables.

2 Kronecker HiPPO-SVGP Inducing Representation

2.1 Temporal HiPPO-RFF Interdomain Features

For the temporal axis, we use HiPPO-LegS interdomain inducing variables together with a random Fourier feature approximation of a stationary temporal kernel.

For a time horizon $T > 0$, define the scaled Legendre basis

$$g_\ell^{(T)}(x) = \sqrt{\frac{2\ell+1}{T}} P_\ell\left(\frac{2x}{T} - 1\right), \quad x \in [0, T], \quad \ell = 0, \dots, M_t - 1, \quad (7)$$

where P_ℓ is the Legendre polynomial of degree ℓ . The temporal interdomain inducing variables are

$$u_\ell^{(t)}(T) = \int_0^T g_\ell^{(T)}(x) f_t(x) dx. \quad (8)$$

Assume the temporal kernel k_t is stationary. By Bochner’s theorem, we approximate it using random Fourier features:

$$k_t(\tau, \tau') \approx \sum_{r=1}^R w_r [\cos(\omega_r \tau) \cos(\omega_r \tau') + \sin(\omega_r \tau) \sin(\omega_r \tau')]. \quad (9)$$

Define the oscillatory integrals

$$I_\ell^{\cos}(T; \omega) = \int_0^T \cos(\omega x) g_\ell^{(T)}(x) dx, \quad (10)$$

$$I_\ell^{\sin}(T; \omega) = \int_0^T \sin(\omega x) g_\ell^{(T)}(x) dx. \quad (11)$$

Using the spherical-Bessel identity

$$\int_{-1}^1 e^{i\kappa u} P_\ell(u) du = 2i^\ell j_\ell(\kappa), \quad \kappa = \frac{\omega T}{2}, \quad (12)$$

these integrals admit closed-form expressions involving spherical Bessel functions $j_\ell(\cdot)$. This gives analytic temporal inducing covariances and point-to-inducing cross-covariances.

Let $B_t(T) \in \mathbb{R}^{M_t \times 2R}$ be the temporal HiPPO-RFF feature matrix, with entries

$$[B_t(T)]_{\ell,r} = \sqrt{w_r} I_\ell^{\cos}(T; \omega_r), \quad [B_t(T)]_{\ell,R+r} = \sqrt{w_r} I_\ell^{\sin}(T; \omega_r), \quad (13)$$

and define the temporal RFF vector

$$\varphi_t(\tau) = (\sqrt{w_1} \cos(\omega_1 \tau), \dots, \sqrt{w_R} \cos(\omega_R \tau), \sqrt{w_1} \sin(\omega_1 \tau), \dots, \sqrt{w_R} \sin(\omega_R \tau))^\top. \quad (14)$$

Then

$$K_{uu}^{(t)}(T) \approx B_t(T) B_t(T)^\top, \quad (15)$$

and

$$k_{fu}^{(t)}(\tau; T) \approx \varphi_t(\tau)^\top B_t(T)^\top. \quad (16)$$

For local temporal mini-batches, the same construction is applied using a local horizon T_n for block D_n , yielding $K_{uu,n}^{(t)}$ and $K_{fu,n}^{(t)}$.

2.2 Spatial Inducing Variables

On the spatial axis, we use inducing locations

$$Z_s = \{z_1^{(s)}, \dots, z_{M_s}^{(s)}\}. \quad (17)$$

The spatial inducing covariance and cross-covariance are

$$K_{ZZ}^{(s)} \in \mathbb{R}^{M_s \times M_s}, \quad [K_{ZZ}^{(s)}]_{jj'} = k_s(z_j^{(s)}, z_{j'}^{(s)}), \quad (18)$$

$$K_{XZ}^{(s)} \in \mathbb{R}^{N_s \times M_s}, \quad [K_{XZ}^{(s)}]_{ij} = k_s(s_i, z_j^{(s)}). \quad (19)$$

The spatial inducing set is fixed across online time steps. This is important: online adaptation occurs through the temporal HiPPO component, while the spatial inducing geometry remains shared across all blocks.

2.3 Mixed Spatio-Temporal Inducing Variables

For each spatial inducing location $z_j^{(s)}$, define temporal interdomain variables

$$u_{\ell,j}(T) = \int_0^T g_{\ell}^{(T)}(x) f(x, z_j^{(s)}) dx. \quad (20)$$

Stacking the temporal and spatial inducing variables gives

$$u(T) = \text{vec}([u_{\ell,j}(T)]_{\ell,j}) \in \mathbb{R}^{M_t M_s}. \quad (21)$$

By separability of the spatio-temporal kernel,

$$K_{uu}^{(st)}(T) = K_{uu}^{(t)}(T) \otimes K_{ZZ}^{(s)}. \quad (22)$$

For a block D_n , the point-to-inducing covariance has the corresponding Kronecker form

$$K_{fu,n}^{(st)} = K_{fu,n}^{(t)} \otimes K_{XZ}^{(s)}. \quad (23)$$

The sparse GP projection matrix for block n is therefore

$$A_n = K_{fu,n}^{(st)} (K_{uu,n}^{(st)})^{-1}. \quad (24)$$

Using Kronecker identities,

$$A_n = \left[K_{fu,n}^{(t)} (K_{uu,n}^{(t)})^{-1} \right] \otimes \left[K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1} \right]. \quad (25)$$

Define

$$T_n := K_{fu,n}^{(t)} (K_{uu,n}^{(t)})^{-1}, \quad C := K_{XZ}^{(s)} (K_{ZZ}^{(s)})^{-1}. \quad (26)$$

Then

$$A_n = T_n \otimes C. \quad (27)$$

3 Why Residual-Based Online Learning is Inconsistent

A two-stage procedure first estimates the linear mean and then fits a GP to residuals:

$$r_n(\hat{\beta}_n) = y_n - \Phi_n \hat{\beta}_n. \quad (28)$$

The difficulty is that residuals are not fixed targets. If the estimate of the linear coefficient changes from $\hat{\beta}_{n-1}$ to $\hat{\beta}_n$, then for any historical block $j < n$,

$$r_j(\hat{\beta}_n) = y_j - \Phi_j \hat{\beta}_n \quad (29)$$

$$= y_j - \Phi_j \hat{\beta}_{n-1} - \Phi_j(\hat{\beta}_n - \hat{\beta}_{n-1}) \quad (30)$$

$$= r_j(\hat{\beta}_{n-1}) - \Phi_j(\hat{\beta}_n - \hat{\beta}_{n-1}). \quad (31)$$

Thus, the historical residual memory becomes stale whenever β is updated.

Proposition 1 (Joint training removes stale residual targets). *In the joint model*

$$y_j \mid \beta, u_j \sim \mathcal{N}(\Phi_j \beta + A_j u_j, \sigma^2 I), \quad (32)$$

historical information is stored as a likelihood contribution to the same joint posterior over (β, u) , rather than as residual observations computed from a changing point estimate of β . Therefore, historical information does not become inconsistent when the posterior mean of β changes.

Proof. The two-stage target for block j is $r_j(\hat{\beta}) = y_j - \Phi_j \hat{\beta}$, which explicitly depends on the current point estimate $\hat{\beta}$. Equation (31) shows that the target changes when $\hat{\beta}$ changes.

By contrast, joint training uses the likelihood

$$p(y_j \mid \beta, u_j) = \mathcal{N}(y_j \mid \Phi_j \beta + A_j u_j, \sigma^2 I). \quad (33)$$

This likelihood is a fixed probabilistic statement about y_j as a function of the unknown variables (β, u_j) . Updating the posterior mean of β changes our belief about β , but it does not redefine the historical observation y_j or the likelihood contribution of block j . Hence no stale residual target is created. $\square$

The essential distinction is

$$\text{two-stage: learn } y - \Phi \hat{\beta}, \quad \text{joint training: learn } p(y \mid \beta, u). \quad (34)$$

This is the reason the rest of this document focuses on a joint online posterior over the linear mean and GP inducing variables.

4 Online Joint Training ELBO

For online block n , the sparse GP approximation gives

$$f_n \mid u_n \approx A_n u_n. \quad (35)$$

Thus the likelihood is

$$p(y_n \mid \beta, u_n) = \mathcal{N}(y_n \mid \Phi_n \beta + A_n u_n, \sigma^2 I). \quad (36)$$

Let the streaming priors entering block n be

$$p_n(\beta) = \mathcal{N}(m_{\beta,n|n-1}, S_{\beta,n|n-1}), \quad (37)$$

$$p_n(u_n) = \mathcal{N}(m_{u,n|n-1}, S_{u,n|n-1}). \quad (38)$$

A general joint Gaussian variational posterior is

$$q_n(\beta, u_n) = \mathcal{N}\left(\begin{bmatrix} \beta \\ u_n \end{bmatrix} \middle| m_n, S_n\right). \quad (39)$$

The online joint training objective is

$$\mathcal{L}_n = \mathbb{E}_{q_n(\beta, u_n)} [\log p(y_n \mid \beta, u_n)] - \text{KL}[q_n(\beta, u_n) \parallel p_n(\beta, u_n)]. \quad (40)$$

A practical factorized version is

$$q_n(\beta, u_n) = q_n(\beta)q_n(u_n), \quad (41)$$

with

$$q_n(\beta) = \mathcal{N}(m_\beta, S_\beta), \quad q_n(u_n) = \mathcal{N}(m_u, S_u). \quad (42)$$

If the streaming prior also factorizes,

$$p_n(\beta, u_n) = p_n(\beta)p_n(u_n), \quad (43)$$

then

$$\mathcal{L}_n = \mathbb{E}_{q_n(\beta)q_n(u_n)} [\log p(y_n \mid \beta, u_n)] \quad (44)$$

$$- \text{KL}[q_n(\beta) \parallel p_n(\beta)] - \text{KL}[q_n(u_n) \parallel p_n(u_n)]. \quad (45)$$

For the Gaussian likelihood,

$$\begin{aligned} \mathbb{E} \log p(y_n \mid \beta, u_n) &= -\frac{B_n}{2} \log(2\pi\sigma^2) \\ &\quad - \frac{1}{2\sigma^2} [\|y_n - \Phi_n m_\beta - A_n m_u\|^2 + \text{tr}(\Phi_n S_\beta \Phi_n^\top) + \text{tr}(A_n S_u A_n^\top)]. \end{aligned} \quad (46)$$

This ELBO is the formal bridge between the joint linear-mean model and sparse variational GP training. When the inducing coordinates are fixed, the previous posterior can be used directly as the next streaming prior. When the HiPPO inducing coordinates change, however, the expression $p_n(u_n)$ must be constructed carefully. The main method in this document uses an SSGP-style old-likelihood-ratio transfer, derived in Section 6. The simpler prior-based ELBO remains useful as a baseline form and as the projected-prior approximation discussed later.

5 Fixed-Basis Exact Joint Gaussian Update

If the inducing representation is fixed across online blocks, i.e. the same inducing variable u is used throughout, then the joint model is conjugate under a Gaussian likelihood. In this special case, the exact joint posterior can be maintained in information form.

Define the joint state

$$z = \begin{bmatrix} \beta \\ u \end{bmatrix}, \quad H_n = \begin{bmatrix} \Phi_n & A_n \end{bmatrix}. \quad (48)$$

Then

$$y_n \mid z \sim \mathcal{N}(H_n z, \sigma^2 I). \quad (49)$$

Let

$$q_{n-1}(z) = \mathcal{N}(m_{n-1}, S_{n-1}), \quad (50)$$

and define the natural parameters

$$\Lambda_{n-1} = S_{n-1}^{-1}, \quad h_{n-1} = S_{n-1}^{-1} m_{n-1}. \quad (51)$$

The exact Gaussian-conjugate update is

$$\Lambda_n = \Lambda_{n-1} + \frac{1}{\sigma^2} H_n^\top H_n, \quad (52)$$

$$h_n = h_{n-1} + \frac{1}{\sigma^2} H_n^\top y_n. \quad (53)$$

The posterior moments are recovered by

$$S_n = \Lambda_n^{-1}, \quad m_n = S_n h_n. \quad (54)$$

Expanding the precision contribution gives

$$H_n^\top H_n = \begin{bmatrix} \Phi_n^\top \Phi_n & \Phi_n^\top A_n \\ A_n^\top \Phi_n & A_n^\top A_n \end{bmatrix}. \quad (55)$$

Therefore,

$$\Lambda_n = \Lambda_{n-1} + \frac{1}{\sigma^2} \begin{bmatrix} \Phi_n^\top \Phi_n & \Phi_n^\top A_n \\ A_n^\top \Phi_n & A_n^\top A_n \end{bmatrix}, \quad (56)$$

and

$$h_n = h_{n-1} + \frac{1}{\sigma^2} \begin{bmatrix} \Phi_n^\top y_n \\ A_n^\top y_n \end{bmatrix}. \quad (57)$$

This update is stronger than a mean-field approximation because it keeps the posterior cross-covariance

$$\text{cov}(\beta, u) \neq 0. \quad (58)$$

The fixed-basis exact joint Gaussian update should therefore be used as a baseline and sanity check for more scalable structured or mean-field approximations.

Remark 1. *Equations (52)–(54) are valid only when all historical likelihood contributions are expressed in the same inducing coordinates. If the HiPPO temporal inducing basis changes over time, the previous posterior must first be transferred to the new inducing coordinates before the next likelihood update is applied.*

6 Changing-Basis Posterior-to-Prior Transfer

In the HiPPO-SVGP setting, the temporal inducing basis depends on the current horizon or on the current local temporal block. Consequently, the inducing variables used at two consecutive online steps are generally not the same random variables. Let u_o denote the old inducing variable at step $n - 1$, and let u_n denote the new inducing variable at step n . In general,

$$u_o \neq u_n. \quad (59)$$

For example, the old and new temporal HiPPO interdomain variables may be defined as

$$u_\ell^{\text{old}} = \int_0^{T_{n-1}} g_\ell^{(T_{n-1})}(x) f(x) dx, \quad u_\ell^{\text{new}} = \int_0^{T_n} g_\ell^{(T_n)}(x) f(x) dx. \quad (60)$$

Because the basis function $g_\ell^{(T)}$ changes with the horizon T , the old posterior $q_{n-1}(u_o)$ cannot be directly reused as a posterior or prior over u_n . Therefore, the naive assignment

$$p_n(u_n) = q_{n-1}(u_o) \quad (61)$$

is not well-defined.

The central problem is thus to transfer the information contained in $q_{n-1}(u_o)$ into the new inducing coordinates u_n . We distinguish two related transfer mechanisms. The first one is a streaming variational free-energy transfer that explicitly propagates the old likelihood information, following the principle of streaming sparse Gaussian processes. This is the main theoretical formulation. The second one is a simpler Gaussian projected-prior transfer, which is useful as an efficient approximation and as an ablation baseline.

6.1 Streaming VFE Transfer with Changing HiPPO Coordinates

The streaming sparse GP framework represents old observations without replaying them. The old data contribution is encoded by the ratio between the previous posterior and the old prior. In the standard notation, let a denote the old inducing variables and let b denote the new inducing variables. The old data contribution is represented through

$$\frac{q_{\text{old}}(a)}{p(a)}. \quad (62)$$

The corresponding variational free-energy update can be written as

$$q_{\text{vfe}}(b) \propto p(b) \exp \left\{ \mathbb{E}_{p(a|b)} \left[\log \frac{q_{\text{old}}(a)}{p(a)} \right] + \mathbb{E}_{p(f_{\text{new}}|b)} [\log p(y_{\text{new}} | f_{\text{new}})] \right\}. \quad (63)$$

The first expectation transfers the old approximate likelihood factor into the new inducing representation, while the second expectation assimilates the likelihood contribution of the new data block.

In our model, the same principle is applied to HiPPO interdomain inducing variables. The old inducing variable is u_o , the new inducing variable is u_n , and the new observation model is the joint linear-mean GP likelihood

$$p(y_n | \beta, u_n) = \mathcal{N}(y_n | \Phi_n \beta + A_n u_n, \sigma^2 I). \quad (64)$$

The direct analogue of the streaming sparse GP VFE update is therefore

$$q_n(u_n) \propto p(u_n) \exp \left\{ \mathbb{E}_{p(u_o|u_n)} \left[\log \frac{q_{n-1}(u_o)}{p(u_o)} \right] + \mathbb{E}_{p(f_n|u_n)} [\log p(y_n | \beta, u_n)] \right\}. \quad (65)$$

The first term transfers the approximate old likelihood information from the old HiPPO basis into the new HiPPO basis. The second term incorporates the new data block under the joint linear-mean GP likelihood.

Equivalently, define the old-information transfer term

$$\mathcal{T}_{o \rightarrow n}(u_n) := \mathbb{E}_{p(u_o|u_n)} \left[\log \frac{q_{n-1}(u_o)}{p(u_o)} \right]. \quad (66)$$

Then

$$q_n(u_n) \propto p(u_n) \exp \left\{ \mathcal{T}_{o \rightarrow n}(u_n) + \mathbb{E}_{p(f_n|u_n)} [\log p(y_n | \beta, u_n)] \right\}. \quad (67)$$

This formulation is stronger than a simple posterior projection because it propagates the old likelihood factor $q_{n-1}(u_o)/p(u_o)$, rather than only the marginal predictive distribution of u_n implied by $q_{n-1}(u_o)$.

6.2 Closed-Form Gaussian Old-Likelihood Transfer

We now derive the closed-form Gaussian version of (65). Assume that the old posterior and old prior are

$$q_{n-1}(u_o) = \mathcal{N}(u_o | m_o, S_o), \quad p(u_o) = \mathcal{N}(u_o | 0, K_{oo}). \quad (68)$$

Then the log old-likelihood ratio is quadratic:

$$\log \frac{q_{n-1}(u_o)}{p(u_o)} = -\frac{1}{2} u_o^\top (S_o^{-1} - K_{oo}^{-1}) u_o + u_o^\top S_o^{-1} m_o + c_o, \quad (69)$$

where c_o collects terms that do not depend on u_o . Define the old likelihood natural parameters

$$R_o := S_o^{-1} - K_{oo}^{-1}, \quad r_o := S_o^{-1} m_o. \quad (70)$$

Thus,

$$\log \frac{q_{n-1}(u_o)}{p(u_o)} = -\frac{1}{2} u_o^\top R_o u_o + u_o^\top r_o + c_o. \quad (71)$$

Under the GP prior, define the joint covariance between the new and old inducing variables by

$$\begin{bmatrix} u_n \\ u_o \end{bmatrix} \sim \mathcal{N} \left(0, \begin{bmatrix} K_{nn} & K_{no} \\ K_{on} & K_{oo} \end{bmatrix} \right). \quad (72)$$

The conditional distribution of the old inducing variable given the new one is

$$p(u_o | u_n) = \mathcal{N}(L_{on}u_n, \Sigma_{o|n}), \quad (73)$$

where

$$L_{on} := K_{on}K_{nn}^{-1}, \quad \Sigma_{o|n} := K_{oo} - K_{on}K_{nn}^{-1}K_{no}. \quad (74)$$

Substituting (71) into (66) gives

$$\mathcal{T}_{o \rightarrow n}(u_n) = \mathbb{E}_{p(u_o|u_n)} \left[-\frac{1}{2}u_o^\top R_o u_o + u_o^\top r_o \right] + \text{const} \quad (75)$$

$$= -\frac{1}{2}u_n^\top L_{on}^\top R_o L_{on} u_n + u_n^\top L_{on}^\top r_o + \text{const}. \quad (76)$$

The trace term $\text{tr}(R_o \Sigma_{o|n})$ is absorbed into the constant because it does not depend on u_n .

Therefore, the old data contribute the following natural parameters in the new inducing coordinates:

$$\Lambda_{o \rightarrow n} = L_{on}^\top R_o L_{on}, \quad h_{o \rightarrow n} = L_{on}^\top r_o. \quad (77)$$

These are the old-likelihood-ratio contributions transferred from the old HiPPO basis to the new HiPPO basis.

For the new data block, under the Gaussian observation model (64), the GP part contributes

$$\Lambda_{\text{new},n} = \frac{1}{\sigma^2} A_n^\top A_n, \quad h_{\text{new},n} = \frac{1}{\sigma^2} A_n^\top (y_n - \Phi_n m_{\beta,n}) \quad (78)$$

when using a mean-field update in which $q_n(\beta)$ and $q_n(u_n)$ are updated alternately.

Combining the GP prior, the transferred old likelihood, and the new likelihood gives

$$S_{u,n}^{-1} = K_{nn}^{-1} + \Lambda_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top A_n, \quad (79)$$

and

$$S_{u,n}^{-1} m_{u,n} = h_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top (y_n - \Phi_n m_{\beta,n}). \quad (80)$$

Equations (79)–(80) are the SSGP-style changing-basis update for the HiPPO inducing variables. They explicitly propagate old likelihood information through the ratio $q_{n-1}(u_o)/p(u_o)$, and therefore more closely follow the streaming sparse GP variational principle than a simple posterior moment projection.

6.3 Joint Linear-Mean Update

The linear coefficient β is updated jointly with the GP inducing state. If β is assumed static, its streaming prior is simply

$$p_n(\beta) = q_{n-1}(\beta). \quad (81)$$

If slowly time-varying linear effects are allowed, we use a drift prior

$$p_n(\beta) = \mathcal{N}(m_{\beta,n-1}, S_{\beta,n-1} + Q_\beta), \quad Q_\beta \succeq 0. \quad (82)$$

For a mean-field posterior

$$q_n(\beta, u_n) = q_n(\beta)q_n(u_n), \quad (83)$$

the coordinate update for β is

$$S_{\beta,n}^{-1} = S_{\beta,n|n-1}^{-1} + \frac{1}{\sigma^2} \Phi_n^\top \Phi_n, \quad (84)$$

and

$$S_{\beta,n}^{-1} m_{\beta,n} = S_{\beta,n|n-1}^{-1} m_{\beta,n|n-1} + \frac{1}{\sigma^2} \Phi_n^\top (y_n - A_n m_{u,n}). \quad (85)$$

The update for u_n is given by (79)–(80). In practice, one alternates (84)–(85) and (79)–(80) for a small number of inner iterations per block.

This joint update is different from a residual-based two-stage method. The likelihood is always applied to the original observations y_n ,

$$p(y_n \mid \beta, u_n) = \mathcal{N}(y_n \mid \Phi_n \beta + A_n u_n, \sigma^2 I), \quad (86)$$

rather than to residuals computed using a changing point estimate of β . Hence historical information is carried through the joint posterior and through the old-likelihood-ratio transfer, not through stale residual targets.

6.4 Projected-Prior Approximation

The SSGP-style transfer above is the main theoretical update. However, it requires constructing and propagating the old-likelihood ratio

$$\log \frac{q_{n-1}(u_o)}{p(u_o)}. \quad (87)$$

For a simpler and cheaper implementation, one may instead use a Gaussian projected-prior approximation. This approximation directly maps the old posterior over u_o into a predictive Gaussian prior over u_n :

$$p_n^{\text{proj}}(u_n) = \int p(u_n | u_o) q_{n-1}(u_o) du_o. \quad (88)$$

Assume again that

$$q_{n-1}(u_o) = \mathcal{N}(m_o, S_o), \quad (89)$$

and use the GP conditional

$$p(u_n | u_o) = \mathcal{N}(K_{no}K_{oo}^{-1}u_o, K_{nn} - K_{no}K_{oo}^{-1}K_{on}). \quad (90)$$

Then

$$p_n^{\text{proj}}(u_n) = \mathcal{N}(m_{u,n|n-1}^{\text{proj}}, S_{u,n|n-1}^{\text{proj}}), \quad (91)$$

where

$$m_{u,n|n-1}^{\text{proj}} = K_{no}K_{oo}^{-1}m_o, \quad (92)$$

and

$$S_{u,n|n-1}^{\text{proj}} = K_{nn} + K_{no}K_{oo}^{-1}(S_o - K_{oo})K_{oo}^{-1}K_{on}. \quad (93)$$

This projected-prior update is a filtering-style moment projection. It preserves the posterior mean and covariance information of the old HiPPO state and maps them into the current basis. If the old posterior equals the old prior, $S_o = K_{oo}$, then

$$S_{u,n|n-1}^{\text{proj}} = K_{nn}, \quad (94)$$

so no information beyond the GP prior is carried forward. Conversely, if $S_o \neq K_{oo}$, then the term $S_o - K_{oo}$ transfers the posterior uncertainty reduction induced by past data.

The projected-prior approximation is easier to implement and is useful as a computationally efficient baseline. However, it does not explicitly propagate the old likelihood ratio $q_{n-1}(u_o)/p(u_o)$. Therefore, the SSGP-style transfer in (79)–(80) is the preferred formulation when the goal is to obtain a more principled streaming variational method.

6.5 Summary of the Changing-Basis Transfer

The changing-basis problem arises because the HiPPO inducing coordinates evolve with the temporal horizon or local block. The old inducing state u_o and the new inducing state u_n are distinct random variables. This section introduced two transfer mechanisms:

- (1) **SSGP-style old-likelihood-ratio transfer.** This is the main theoretical update. It propagates the old approximate likelihood factor $q_{n-1}(u_o)/p(u_o)$ through $p(u_o | u_n)$, giving the transferred natural parameters

$$\Lambda_{o \rightarrow n} = L_{on}^\top R_o L_{on}, \quad h_{o \rightarrow n} = L_{on}^\top r_o. \quad (95)$$

The resulting posterior update is

$$S_{u,n}^{-1} = K_{nn}^{-1} + \Lambda_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top A_n. \quad (96)$$

- (2) **Gaussian projected-prior transfer.** This is a simpler filtering-style approximation:

$$p_n^{\text{proj}}(u_n) = \int p(u_n | u_o) q_{n-1}(u_o) du_o. \quad (97)$$

It is computationally convenient but less directly tied to the streaming sparse GP variational free-energy derivation.

In the remainder of the method, the SSGP-style transfer is treated as the main principled changing-basis update, while the projected-prior transfer serves as an efficient approximation and an ablation baseline. This separation allows the model to retain a strong connection to streaming variational sparse GP inference while still supporting practical Kronecker-structured HiPPO-SVGP implementation.

7 Mean-Field Coordinate Updates

The full joint posterior over (β, u_n) is accurate but may be expensive when $M = M_t M_s$ is large. A scalable engineering version is the factorized posterior

$$q_n(\beta, u_n) = q_n(\beta) q_n(u_n). \quad (98)$$

This section gives the coordinate updates consistent with the SSGP-style transfer in Section 6. The projected-prior approximation is also included as an implementation baseline.

7.1 Coordinate Update for the Linear Coefficient

Let the streaming prior for β entering block n be

$$p_n(\beta) = \mathcal{N}(m_{\beta,n|n-1}, S_{\beta,n|n-1}). \quad (99)$$

Holding $q_n(u_n)$ fixed, the coordinate-optimal Gaussian factor for β is

$$S_{\beta,n}^{-1} = S_{\beta,n|n-1}^{-1} + \frac{1}{\sigma^2} \Phi_n^\top \Phi_n, \quad (100)$$

and

$$S_{\beta,n}^{-1} m_{\beta,n} = S_{\beta,n|n-1}^{-1} m_{\beta,n|n-1} + \frac{1}{\sigma^2} \Phi_n^\top (y_n - A_n m_{u,n}). \quad (101)$$

This resembles a residual update, but it is only an inner coordinate step of one joint ELBO. No historical residual target is stored.

7.2 Coordinate Update for the GP Inducing State under SSGP-Style Transfer

Holding $q_n(\beta)$ fixed, the coordinate-optimal Gaussian factor for u_n uses three sources of information: the current GP prior $p(u_n) = \mathcal{N}(0, K_{nn})$, the transferred old-likelihood factor, and the new block likelihood. Therefore,

$$S_{u,n}^{-1} = K_{nn}^{-1} + \Lambda_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top A_n, \quad (102)$$

and

$$S_{u,n}^{-1} m_{u,n} = h_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top (y_n - \Phi_n m_{\beta,n}). \quad (103)$$

Here

$$\Lambda_{o \rightarrow n} = L_{on}^\top R_o L_{on}, \quad h_{o \rightarrow n} = L_{on}^\top r_o, \quad (104)$$

with

$$R_o = S_o^{-1} - K_{oo}^{-1}, \quad r_o = S_o^{-1} m_o. \quad (105)$$

Equations (102)–(103) are the mean-field coordinate update corresponding to the SSGP-style old-likelihood-ratio transfer.

In practice, one performs a small number of inner sweeps per online block:

- (i) initialize $q_n(\beta)$ from $p_n(\beta)$;
- (ii) construct the old-likelihood transfer $(\Lambda_{o \rightarrow n}, h_{o \rightarrow n})$;
- (iii) update $q_n(u_n)$ using (102)–(103);
- (iv) update $q_n(\beta)$ using (100)–(101);
- (v) repeat the last two steps if needed.

7.3 Projected-Prior Coordinate Update

For comparison, the projected-prior approximation replaces the transferred old-likelihood factor by a Gaussian streaming prior

$$p_n^{\text{proj}}(u_n) = \mathcal{N}\left(m_{u,n|n-1}^{\text{proj}}, S_{u,n|n-1}^{\text{proj}}\right). \quad (106)$$

In that approximation,

$$S_{u,n}^{-1} = \left(S_{u,n|n-1}^{\text{proj}}\right)^{-1} + \frac{1}{\sigma^2} A_n^\top A_n, \quad (107)$$

and

$$S_{u,n}^{-1} m_{u,n} = \left(S_{u,n|n-1}^{\text{proj}}\right)^{-1} m_{u,n|n-1}^{\text{proj}} + \frac{1}{\sigma^2} A_n^\top (y_n - \Phi_n m_{\beta,n}). \quad (108)$$

This projected-prior version is used as an efficient approximation or ablation baseline, not as the main theoretical update.

8 Kronecker-Aware Implementation

The Kronecker form

$$A_n = T_n \otimes C \quad (109)$$

should be used explicitly. Therefore,

$$A_n^\top A_n = (T_n^\top T_n) \otimes (C^\top C). \quad (110)$$

Let

$$G := C^\top C. \quad (111)$$

Since the spatial inducing set is fixed, G can be precomputed.

Under the SSGP-style changing-basis transfer, the GP inducing precision contains three terms:

$$\Lambda_{u,n} = K_{nn}^{-1} + \Lambda_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top A_n. \quad (112)$$

The main challenge is the transferred old-likelihood precision $\Lambda_{o \rightarrow n} = L_{on}^\top R_o L_{on}$. In a generic model this term need not be Kronecker-separable. In the present construction, however, it is Kronecker-preserving because the inducing basis changes only along the temporal dimension, while the spatial inducing geometry remains fixed.

8.1 Kronecker-Preserving Old-Likelihood Transfer

Let

$$K_{nn} = K_{nn}^{(t)} \otimes K_s, \quad K_{oo} = K_{oo}^{(t)} \otimes K_s, \quad K_{on} = K_{on}^{(t)} \otimes K_s, \quad (113)$$

where $K_s = K_{ZZ}^{(s)}$. Since the spatial inducing locations are fixed, the spatial factor is the same in the old, new, and cross covariances. Then

$$L_{on} = K_{on} K_{nn}^{-1} \quad (114)$$

$$= (K_{on}^{(t)} \otimes K_s) \left[(K_{nn}^{(t)})^{-1} \otimes K_s^{-1} \right] \quad (115)$$

$$= \left[K_{on}^{(t)} (K_{nn}^{(t)})^{-1} \right] \otimes I_s. \quad (116)$$

Define

$$L_{on}^{(t)} := K_{on}^{(t)} (K_{nn}^{(t)})^{-1}. \quad (117)$$

Then

$$L_{on} = L_{on}^{(t)} \otimes I_s. \quad (118)$$

Moreover, because each block likelihood projection has the form

$$A_j = T_j \otimes C, \quad (119)$$

each Gaussian likelihood contribution to the inducing precision is

$$\frac{1}{\sigma^2} A_j^\top A_j = \frac{1}{\sigma^2} (T_j^\top T_j) \otimes G. \quad (120)$$

We therefore maintain the old likelihood precision directly in Kronecker natural form,

$$R_o = B_o \otimes G, \quad (121)$$

rather than recovering it by dense subtraction $R_o = S_o^{-1} - K_{oo}^{-1}$. This is important: forming a dense posterior covariance and then subtracting the prior precision may destroy the structure numerically, while the natural likelihood statistic remains structured by construction.

Substituting (118) and (121) into the old-likelihood transfer gives

$$\Lambda_{o \rightarrow n} = L_{on}^\top R_o L_{on} \quad (122)$$

$$= \left[(L_{on}^{(t)})^\top \otimes I_s \right] (B_o \otimes G) \left[L_{on}^{(t)} \otimes I_s \right] \quad (123)$$

$$= \left[(L_{on}^{(t)})^\top B_o L_{on}^{(t)} \right] \otimes G. \quad (124)$$

Hence the transferred old likelihood remains Kronecker-separable. Define

$$B_{o \rightarrow n} := (L_{on}^{(t)})^\top B_o L_{on}^{(t)}. \quad (125)$$

Then

$$\Lambda_{o \rightarrow n} = B_{o \rightarrow n} \otimes G. \quad (126)$$

The new block likelihood contributes

$$\frac{1}{\sigma^2} A_n^\top A_n = \frac{1}{\sigma^2} (T_n^\top T_n) \otimes G. \quad (127)$$

Therefore the temporal likelihood statistic is updated by

$$B_n = B_{o \rightarrow n} + \frac{1}{\sigma^2} T_n^\top T_n. \quad (128)$$

The resulting inducing precision is

$$\Lambda_{u,n} = (K_{nn}^{(t)})^{-1} \otimes K_s^{-1} + B_n \otimes G. \quad (129)$$

Thus, the SSGP-style old-likelihood-ratio transfer does not destroy the Sylvester structure, provided that the spatial inducing set is fixed and the changing basis acts only along the temporal dimension.

Remark 2. *If the spatial observation pattern or spatial projection changes across online blocks, then the spatial factor G may become block-dependent. The precision then has the more general form*

$$K_t^{-1} \otimes K_s^{-1} + \sum_j B_j \otimes G_j,$$

which no longer reduces to the single Sylvester equation below. It can still be handled by Kronecker matrix-vector products and iterative solvers, but the closed-form Sylvester diagonalization is exact only when the spatial factor is shared or approximated by a shared structured factor.

8.2 Information Vector Transfer

The old information vector is transferred by

$$h_{o \rightarrow n} = L_{on}^\top r_o = \left[(L_{on}^{(t)})^\top \otimes I_s \right] r_o. \quad (130)$$

This multiplication should be implemented as a Kronecker matrix-vector product. If $r_o = \text{vec}(R_{\text{info},o})$, the transformation corresponds to multiplying the reshaped information

matrix on the temporal side; the exact left/right convention depends only on the chosen vectorization order.

The new block information vector is

$$h_{\text{new},n} = \frac{1}{\sigma^2} A_n^\top (y_n - \Phi_n m_{\beta,n}), \quad (131)$$

and it should also be computed using $A_n = T_n \otimes C$, without materializing the full matrix A_n .

8.3 Sylvester Reformulation for Posterior Mean and Uncertainty

With the Kronecker-preserving transfer in (129), linear systems involving $\Lambda_{u,n}$ can still be reduced to Sylvester equations. This applies both to the posterior mean solve and to predictive uncertainty quadratic forms.

Let

$$q = \text{vec}(Q), \quad z = \Lambda_{u,n}^{-1} q = \text{vec}(Z), \quad (132)$$

where $Q, Z \in \mathbb{R}^{M_s \times M_t}$. Using

$$\Lambda_{u,n} = (K_{nn}^{(t)})^{-1} \otimes K_s^{-1} + B_n \otimes G, \quad (133)$$

and the identity

$$(A \otimes B) \text{vec}(Z) = \text{vec}(BZA^\top), \quad (134)$$

the linear system

$$\Lambda_{u,n} z = q \quad (135)$$

is equivalent to

$$K_s^{-1} Z (K_{nn}^{(t)})^{-1} + GZB_n = Q, \quad (136)$$

assuming $K_{nn}^{(t)}$ and B_n are symmetric. After solving for Z ,

$$q^\top \Lambda_{u,n}^{-1} q = \langle Q, Z \rangle_F. \quad (137)$$

Thus, the same Sylvester mechanism can be used after SSGP-style transfer. The key is to maintain the old likelihood precision as the temporal statistic B_o rather than as a dense matrix.

9 Prediction and Uncertainty

For a test point $x_* = (t_*, s_*)$, let

$$\phi_* = \phi(t_*, s_*), \quad (138)$$

and define

$$a_*^\top = K_{*u} K_{uu}^{-1}. \quad (139)$$

In the main method, the posterior moments m_u and S_u used below are obtained from the SSGP-style old-likelihood-ratio transfer update in (79)–(80). When the projected-prior approximation is used as an ablation, the same predictive equations apply, but m_u and S_u are produced by the projected-prior coordinate update.

Under the mean-field posterior $q(\beta)q(u)$, the predictive mean is

$$\mathbb{E}[y_*] = \phi_*^\top m_\beta + a_*^\top m_u. \quad (140)$$

The predictive variance is

$$\text{Var}(y_*) = \sigma^2 + \phi_*^\top S_\beta \phi_* + k(x_*, x_*) - K_{*u} K_{uu}^{-1} K_{u*} + a_*^\top S_u a_*. \quad (141)$$

The five terms correspond to observation noise, linear-mean posterior uncertainty, prior GP variance, inducing conditioning variance reduction, and posterior uncertainty in the GP inducing variables. The last term can be evaluated by the Sylvester solve in Section 8.3.

Under a full joint posterior

$$q(z) = \mathcal{N}(m_z, S_z), \quad z = \begin{bmatrix} \beta \\ u \end{bmatrix}, \quad (142)$$

define

$$h_* = \begin{bmatrix} \phi_* \\ a_* \end{bmatrix}. \quad (143)$$

The predictive mean is

$$\mathbb{E}[y_*] = h_*^\top m_z, \quad (144)$$

and the predictive variance is

$$\text{Var}(y_*) = \sigma^2 + \text{Var}(f_* | u) + h_*^\top S_z h_*. \quad (145)$$

If

$$S_z = \begin{bmatrix} S_{\beta\beta} & S_{\beta u} \\ S_{u\beta} & S_{uu} \end{bmatrix}, \quad (146)$$

then

$$h_*^\top S_z h_* = \phi_*^\top S_{\beta\beta} \phi_* + a_*^\top S_{uu} a_* + 2\phi_*^\top S_{\beta u} a_*. \quad (147)$$

The cross term $2\phi_*^\top S_{\beta u} a_*$ is absent in the mean-field approximation. This is why the fixed-basis exact joint Gaussian update remains useful as a calibration baseline.

10 Experiments and Ablations

The recommended experimental sequence is designed to isolate the contribution of each modeling choice: joint training, changing-basis transfer, old-likelihood-ratio transfer, Kronecker structure, and posterior factorization.

10.1 Methods to Compare

At minimum, compare the following methods:

- (1) **Two-stage future-only baseline.** The linear mean is updated first and the GP is trained on current residuals only.
- (2) **Two-stage residual GP with buffer or sketch correction.** This tests whether explicit residual repair can mitigate stale residual targets.
- (3) **Fixed-basis exact joint Gaussian baseline.** This uses a fixed inducing representation and performs the exact joint Gaussian update over (β, u) .
- (4) **Joint mean-field without changing-basis transfer.** This tests the failure mode caused by ignoring that u_{n-1} and u_n live in different coordinates.
- (5) **Joint mean-field with Gaussian projected-prior transfer.** This is the efficient filtering-style approximation.
- (6) **Joint mean-field with SSGP-style old-likelihood-ratio transfer.** This is the main proposed streaming variational method.
- (7) **Structured joint posterior with SSGP-style transfer.** This tests whether retaining part of the β - u posterior dependence improves calibration.

10.2 Ablations

Important ablations include:

- (1) number of temporal HiPPO basis functions M_t ;
- (2) number of spatial inducing points M_s ;
- (3) number of RFF frequencies R ;
- (4) fixed temporal basis versus changing HiPPO basis;
- (5) no transfer versus projected-prior transfer versus old-likelihood-ratio transfer;
- (6) exact old-likelihood transfer versus structured approximation to $\Lambda_{o \rightarrow n}$;
- (7) maintaining R_o as a dense matrix versus maintaining $B_o \otimes G$;
- (8) mean-field posterior versus full or structured joint posterior;
- (9) static β versus drifting β with Q_β ;
- (10) PSD projection, damping, or jitter for $R_o = S_o^{-1} - K_{oo}^{-1}$ when a dense approximation is used.

10.3 Metrics

The evaluation should not rely only on RMSE. Report:

- (1) RMSE;
- (2) MAE;
- (3) test negative log likelihood;
- (4) predictive interval coverage, for example 50%, 90%, and 95%;
- (5) runtime per online block;
- (6) peak memory;
- (7) RMSE/NLL under matched memory budgets;
- (8) calibration error for Gaussian predictive intervals;
- (9) error on continual evaluation windows, not only future-only forecasting.

The most important comparison is whether SSGP-style joint training improves continual evaluation and uncertainty calibration relative to residual-based two-stage learning and relative to the simpler projected-prior transfer.

11 Summary

The proposed model follows the logical structure

$$\begin{aligned}
& \text{two-stage learning has stale residuals} \\
& \Rightarrow \text{joint training fixes the observation target} \\
& \Rightarrow \text{SVGP ELBO gives the variational framework} \\
& \Rightarrow \text{HiPPO changing basis requires old-information transfer} \\
& \Rightarrow \text{SSGP-style old-likelihood-ratio transfer gives a principled streaming update} \\
& \Rightarrow \text{Kronecker structure makes the method scalable.}
\end{aligned} \tag{148}$$

The final online observation model is

$$y_n = \Phi_n \beta + A_n u_n + \epsilon_n, \quad \epsilon_n \sim \mathcal{N}(0, \sigma^2 I). \tag{149}$$

The key distinction from residual-based learning is that the likelihood is always applied to y_n , not to a residual target that changes whenever β is updated:

$$p(y_n \mid \beta, u_n) = \mathcal{N}(y_n \mid \Phi_n \beta + A_n u_n, \sigma^2 I). \tag{150}$$

For the linear coefficient, the streaming prior is

$$p_n(\beta) = q_{n-1}(\beta), \tag{151}$$

or, with slow drift,

$$p_n(\beta) = \mathcal{N}(m_{\beta, n-1}, S_{\beta, n-1} + Q_\beta). \tag{152}$$

For the GP inducing variables under a changing HiPPO temporal basis, the main proposed update follows the streaming sparse GP principle. Instead of directly reusing $q_{n-1}(u_o)$, we propagate the old likelihood ratio

$$\frac{q_{n-1}(u_o)}{p(u_o)}. \tag{153}$$

For Gaussian $q_{n-1}(u_o) = \mathcal{N}(m_o, S_o)$ and $p(u_o) = \mathcal{N}(0, K_{oo})$, define

$$R_o = S_o^{-1} - K_{oo}^{-1}, \quad r_o = S_o^{-1} m_o. \tag{154}$$

Using

$$p(u_o \mid u_n) = \mathcal{N}(L_{on} u_n, \Sigma_{o|n}), \quad L_{on} = K_{on} K_{nn}^{-1}, \tag{155}$$

the old likelihood information is transferred to the new HiPPO inducing coordinates through

$$\Lambda_{o \rightarrow n} = L_{on}^\top R_o L_{on}, \quad h_{o \rightarrow n} = L_{on}^\top r_o. \quad (156)$$

The main GP coordinate update is therefore

$$S_{u,n}^{-1} = K_{nn}^{-1} + \Lambda_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top A_n, \quad (157)$$

and

$$S_{u,n}^{-1} m_{u,n} = h_{o \rightarrow n} + \frac{1}{\sigma^2} A_n^\top (y_n - \Phi_n m_{\beta,n}). \quad (158)$$

Crucially, this SSGP-style transfer can preserve the Sylvester structure under the Kronecker-separable HiPPO-SVGP assumptions. Since the changing basis acts only in the temporal dimension and the spatial inducing set is fixed,

$$L_{on} = L_{on}^{(t)} \otimes I_s. \quad (159)$$

If the old likelihood precision is maintained as

$$R_o = B_o \otimes G, \quad G = C^\top C, \quad (160)$$

then

$$\Lambda_{o \rightarrow n} = \left[(L_{on}^{(t)})^\top B_o L_{on}^{(t)} \right] \otimes G. \quad (161)$$

Thus the updated precision keeps the form

$$\Lambda_{u,n} = (K_{nn}^{(t)})^{-1} \otimes K_s^{-1} + B_n \otimes G, \quad (162)$$

with

$$B_n = (L_{on}^{(t)})^\top B_o L_{on}^{(t)} + \frac{1}{\sigma^2} T_n^\top T_n. \quad (163)$$

Therefore, the Sylvester solver remains applicable:

$$K_s^{-1} Z (K_{nn}^{(t)})^{-1} + G Z B_n = Q. \quad (164)$$

The Gaussian projected-prior update

$$p_n^{\text{proj}}(u_n) = \int p(u_n \mid u_o) q_{n-1}(u_o) du_o \quad (165)$$

is retained as an efficient approximation and ablation baseline. The main theoretical contribution, however, is the SSGP-style old-likelihood-ratio transfer together with the Kronecker-preserving temporal transfer statistic. This combination gives a principled

streaming variational method for joint linear-mean HiPPO-SVGP while preserving the scalable Sylvester computation required for large spatio-temporal inducing states.

A Proofs for Kronecker-Preserving Changing-Basis Transfer

This appendix gives the detailed proof of the two structural claims used in the main text:

$$L_{on} = L_{on}^{(t)} \otimes I_s, \quad (166)$$

and

$$R_o = B_o \otimes G. \quad (167)$$

The first identity is a consequence of the separable spatio-temporal kernel and the fact that the spatial inducing set is fixed. The second identity is not true for an arbitrary dense posterior covariance; it is an algorithmic invariant obtained by maintaining the old likelihood natural precision in a Kronecker-structured form.

A.1 Proof that $L_{on} = L_{on}^{(t)} \otimes I_s$

Assume the spatio-temporal kernel is separable:

$$k((t, s), (t', s')) = k_t(t, t')k_s(s, s'). \quad (168)$$

Let the mixed HiPPO interdomain inducing variables be

$$u_{\ell,j}(T) = \int_0^T g_{\ell}^{(T)}(x) f(x, z_j^{(s)}) dx, \quad (169)$$

where ℓ indexes the temporal HiPPO basis and j indexes the fixed spatial inducing location $z_j^{(s)}$.

Let

$$u_o := u(T_o), \quad u_n := u(T_n), \quad (170)$$

where $T_o = T_{n-1}$ is the old temporal horizon and T_n is the new temporal horizon. The temporal HiPPO basis changes from $g_{\ell}^{(T_o)}$ to $g_{\ell}^{(T_n)}$, but the spatial inducing locations $Z_s = \{z_1^{(s)}, \dots, z_{M_s}^{(s)}\}$ are fixed.

For two inducing entries,

$$u_{\ell,j}^{old} = \int_0^{T_o} g_{\ell}^{(T_o)}(x) f(x, z_j^{(s)}) dx, \quad (171)$$

and

$$u_{m,j'}^{new} = \int_0^{T_n} g_m^{(T_n)}(x') f(x', z_{j'}^{(s)}) dx', \quad (172)$$

their prior covariance is

$$\text{cov}(u_{\ell,j}^{old}, u_{m,j'}^{new}) = \int_0^{T_o} \int_0^{T_n} g_{\ell}^{(T_o)}(x) g_m^{(T_n)}(x') k((x, z_j^{(s)}), (x', z_{j'}^{(s)})) dx' dx \quad (173)$$

$$= \int_0^{T_o} \int_0^{T_n} g_{\ell}^{(T_o)}(x) g_m^{(T_n)}(x') k_t(x, x') k_s(z_j^{(s)}, z_{j'}^{(s)}) dx' dx \quad (174)$$

$$= \left[\int_0^{T_o} \int_0^{T_n} g_{\ell}^{(T_o)}(x) g_m^{(T_n)}(x') k_t(x, x') dx' dx \right] k_s(z_j^{(s)}, z_{j'}^{(s)}). \quad (175)$$

Thus the old-new covariance factorizes into a temporal covariance factor and a spatial covariance factor. With the vectorization convention used in the main text,

$$K_{on} = K_{on}^{(t)} \otimes K_s, \quad K_s := K_{ZZ}^{(s)}. \quad (176)$$

Similarly,

$$K_{nn} = K_{nn}^{(t)} \otimes K_s, \quad K_{oo} = K_{oo}^{(t)} \otimes K_s. \quad (177)$$

The conditional mean operator in

$$p(u_o | u_n) = \mathcal{N}(L_{on} u_n, \Sigma_{o|n}) \quad (178)$$

is

$$L_{on} = K_{on} K_{nn}^{-1}. \quad (179)$$

Using (176)–(177) and the Kronecker identities

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}, \quad (A \otimes B)(C \otimes D) = (AC) \otimes (BD), \quad (180)$$

we obtain

$$L_{on} = (K_{on}^{(t)} \otimes K_s) \left[(K_{nn}^{(t)})^{-1} \otimes K_s^{-1} \right] \quad (181)$$

$$= K_{on}^{(t)} (K_{nn}^{(t)})^{-1} \otimes K_s K_s^{-1} \quad (182)$$

$$= L_{on}^{(t)} \otimes I_s, \quad (183)$$

where

$$L_{on}^{(t)} := K_{on}^{(t)} \left(K_{nn}^{(t)} \right)^{-1}. \quad (184)$$

Therefore, under a separable kernel, fixed spatial inducing locations, and temporal-only changing basis, the old-to-new conditional operator acts only on the temporal inducing coordinates and leaves the spatial inducing coordinates unchanged.

Vectorization convention. The identity (183) assumes the same vectorization ordering as the Kronecker convention $K_{uu} = K_{uu}^{(t)} \otimes K_s$. If an implementation uses the opposite vectorization order, the equivalent identity becomes $L_{on} = I_s \otimes L_{on}^{(t)}$. The mathematics is unchanged, but the code must be consistent with the chosen ordering.

A.2 Proof and Status of $R_o = B_o \otimes G$

The identity

$$R_o = B_o \otimes G \quad (185)$$

should be interpreted carefully. It is not a property of an arbitrary dense posterior covariance. It is a structured natural-parameter representation of the accumulated old likelihood information.

For a Gaussian observation model, the likelihood contribution of a block j to the inducing-variable precision is

$$\Lambda_{\text{like},j} = \frac{1}{\sigma^2} A_j^\top A_j. \quad (186)$$

Under the Kronecker HiPPO-SVGP representation,

$$A_j = T_j \otimes C, \quad (187)$$

where T_j is the temporal projection matrix for block j , and

$$C = K_{XZ}^{(s)} \left(K_{ZZ}^{(s)} \right)^{-1} \quad (188)$$

is the fixed spatial projection matrix. Therefore,

$$\Lambda_{\text{like},j} = \frac{1}{\sigma^2} (T_j \otimes C)^\top (T_j \otimes C) \quad (189)$$

$$= \frac{1}{\sigma^2} (T_j^\top T_j) \otimes (C^\top C). \quad (190)$$

Define

$$G := C^\top C. \quad (191)$$

Then each block contributes

$$\Lambda_{\text{like},j} = \frac{1}{\sigma^2}(T_j^\top T_j) \otimes G. \quad (192)$$

The old likelihood natural precision before processing block n is the sum of historical likelihood precision contributions, expressed in the old inducing coordinates:

$$R_o := \sum_{j < n} \Lambda_{\text{like},j}. \quad (193)$$

If the spatial projection matrix C is fixed across the online blocks, then the spatial factor $G = C^\top C$ is common to all terms. Hence

$$R_o = \sum_{j < n} \frac{1}{\sigma^2}(T_j^\top T_j) \otimes G \quad (194)$$

$$= \left[\sum_{j < n} \frac{1}{\sigma^2} T_j^\top T_j \right] \otimes G. \quad (195)$$

Define the temporal old-likelihood precision statistic

$$B_o := \sum_{j < n} \frac{1}{\sigma^2} T_j^\top T_j. \quad (196)$$

Then

$$R_o = B_o \otimes G. \quad (197)$$

Thus $R_o = B_o \otimes G$ is valid when the algorithm maintains the old likelihood precision directly in natural-parameter form. In exact Gaussian natural-parameter inference, this agrees with

$$R_o = S_o^{-1} - K_{oo}^{-1}, \quad (198)$$

provided S_o^{-1} denotes the corresponding inducing-variable posterior precision in the same structured representation. However, if one stores an unrestricted dense posterior covariance, a marginal covariance from a full joint posterior over (β, u_o) , or an arbitrary structured approximation, then the dense quantity $S_o^{-1} - K_{oo}^{-1}$ need not be a single Kronecker product.

A.3 Kronecker Preservation of the Old-Likelihood Transfer

Combining the previous two results proves the Kronecker preservation of the SSGP-style old-likelihood-ratio transfer.

The transferred old likelihood precision is

$$\Lambda_{o \rightarrow n} = L_{on}^\top R_o L_{on}. \quad (199)$$

Using

$$L_{on} = L_{on}^{(t)} \otimes I_s, \quad R_o = B_o \otimes G, \quad (200)$$

we have

$$\Lambda_{o \rightarrow n} = \left[(L_{on}^{(t)})^\top \otimes I_s \right] (B_o \otimes G) [L_{on}^{(t)} \otimes I_s] \quad (201)$$

$$= \left[(L_{on}^{(t)})^\top B_o L_{on}^{(t)} \right] \otimes G. \quad (202)$$

Define

$$B_{o \rightarrow n} := (L_{on}^{(t)})^\top B_o L_{on}^{(t)}. \quad (203)$$

Then

$$\Lambda_{o \rightarrow n} = B_{o \rightarrow n} \otimes G. \quad (204)$$

The new block likelihood contribution is

$$\frac{1}{\sigma^2} A_n^\top A_n = \frac{1}{\sigma^2} (T_n^\top T_n) \otimes G. \quad (205)$$

Therefore the updated temporal likelihood statistic is

$$B_n = B_{o \rightarrow n} + \frac{1}{\sigma^2} T_n^\top T_n. \quad (206)$$

The resulting inducing precision has the Sylvester-compatible form

$$\Lambda_{u,n} = (K_{nn}^{(t)})^{-1} \otimes K_s^{-1} + B_n \otimes G. \quad (207)$$

Consequently, for $q_* = \text{vec}(Q)$ and $z = \Lambda_{u,n}^{-1} q_* = \text{vec}(Z)$, the system

$$\Lambda_{u,n} z = q_* \quad (208)$$

is equivalent to the Sylvester-type equation

$$K_s^{-1} Z (K_{nn}^{(t)})^{-1} + G Z B_n = Q, \quad (209)$$

assuming the vectorization convention used in the main text and symmetric B_n . Thus the SSGP-style old-likelihood-ratio transfer can preserve the Sylvester solver under the temporal-only changing-basis and fixed spatial-geometry assumptions.

A.4 When $R_o = B_o \otimes G$ Can Fail

The single-Kronecker representation $R_o = B_o \otimes G$ is not unconditional. It may fail in the following cases.

Changing spatial observation pattern. If the spatial observation pattern or spatial mini-batch changes across blocks, then the spatial projection matrix may become block-dependent:

$$C_j \neq C. \quad (210)$$

Consequently,

$$G_j = C_j^\top C_j \quad (211)$$

also changes. The old likelihood precision becomes

$$R_o = \sum_{j < n} B_j \otimes G_j, \quad (212)$$

rather than a single Kronecker product $B_o \otimes G$. This is still a sum-of-Kronecker structure, but the simple closed-form Sylvester equation in (209) no longer applies directly. One would need a generalized Sylvester solver or an iterative method using Kronecker matrix-vector products.

Changing spatial inducing locations. If the spatial inducing locations Z_s are updated online, then

$$K_s = K_{ZZ}^{(s)}, \quad C = K_{XZ}^{(s)} \left(K_{ZZ}^{(s)} \right)^{-1}, \quad G = C^\top C \quad (213)$$

all change over time. In that case, the old-to-new transfer operator need not have the form $L_{on}^{(t)} \otimes I_s$, and the likelihood precision is generally not a single $B_o \otimes G$ term.

Recovering R_o from a dense posterior covariance. If one stores a full joint posterior over (β, u_o) and then uses only the marginal covariance of u_o , or if one stores an unrestricted dense posterior covariance S_o and directly computes

$$R_o = S_o^{-1} - K_{oo}^{-1}, \quad (214)$$

then R_o is generally not a single Kronecker product unless the posterior precision itself preserves the required structure. For the Sylvester-preserving implementation, the old likelihood natural precision should be maintained directly as $B_o \otimes G$, rather than recovered from an arbitrary dense covariance.

Non-Gaussian likelihoods or non-conjugate approximations. For Gaussian likelihoods, the likelihood precision contribution is

$$\frac{1}{\sigma^2} A_j^\top A_j. \quad (215)$$

With non-Gaussian likelihoods, approximate inference methods such as Laplace, expectation propagation, or stochastic variational inference may produce likelihood-site precisions that are not simply proportional to $A_j^\top A_j$, or that vary by observation. The resulting precision may fail to share a common spatial factor G .

Practical implication. To retain the simple Sylvester solver, the scalable version of the method should explicitly assume fixed spatial inducing geometry and should maintain the old likelihood precision as a Kronecker natural statistic:

$$R_o = B_o \otimes G. \quad (216)$$

When these assumptions are violated, the model can still be implemented, but the solver must be replaced by a more general structured linear algebra routine, such as a sum-of-Kronecker conjugate-gradient method or a generalized Sylvester solver.